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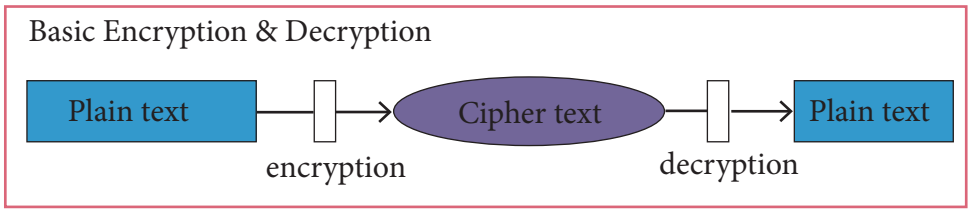

Figure 17.9 Encryption and Decryption

Encryption has been used by militaries and governments to facilitate secret communication. It is now commonly used in protecting information within many kinds of civilian systems. It is also used to protect data in communication system, for example data being transferred via networks (e.g. the Internet, ecommerce), mobile telephones, wireless microphones, wireless intercom systems, Bluetooth devices and bank automatic teller machines. There have been numerous reports of data in communication being intercepted in recent years. Data should also be encrypted when transmitted across networks in order to protect against the network traffic by unauthorized users.

TYPES OF ENCRYPTION

There are two types of encryption schemes as listed below:

- Symmetric Key encryption
- Public Key encryption

SYMMETRIC KEY ENCRYPTION

Symmetric encryption is a technique to use the same key for both encryption and decryption. The main disadvantage of the symmetric key encryption is that all authorized persons involved, have to exchange the key used to encrypt the data before they can decrypt it. If anybody intercepts the key information, they may read all message. **Figure 17.10** depicts the working of symmetric key encryption.

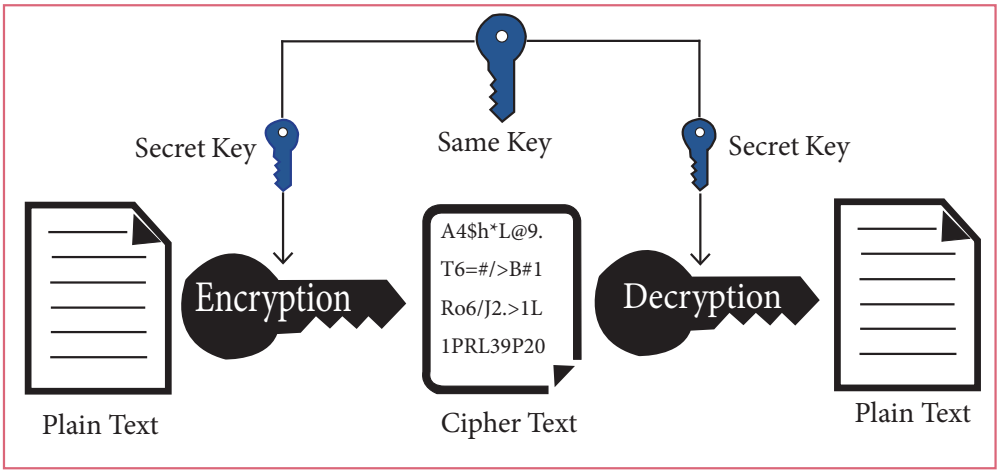

Figure 17.10 Symmetric key encryption